main.py ×

```python
angles = range(0, 360, 10)
image_filenames = [f"earth_{angle}_degrees.jpg" for angle in angles]

resized_earth_images = []
for filename in image_filenames:
    img = Image.open(filename)
    resized_img = img.resize((256, 256), Image.ANTIALIAS)
    resized_earth_images.append(resized_img)

output_gif = "rotating_earth.gif"
resized_earth_images[0].save(
    output_gif,
    save_all=True,
    append_images=resized_earth_images[1:],
    duration=100,
    loop=0,
    optimize = True
)

from PIL import ImageChops

def crossfade(image1, image2, alpha):
    return ImageChops.blend(image1, image2, alpha)

crossfade_frames = []
for i in range(len(resized_earth_images) - 1):
    for alpha in (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9):
        frame = crossfade(resized_earth_images[i], resized_earth_images[i + 1], alpha)
        crossfade_frames.append(frame)
```

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Guide ×

Collapse

Adding Transitions and Visual Effects

In this section, we will explore how to enhance the GIF animation by adding transitions and visual effects to the images generated by the DALL-E 2 API. Adding transitions can make the animation smoother, while visual effects can create a more engaging and dynamic presentation.

1. **Adding Crossfade Transitions**:

A crossfade transition blends two consecutive images, gradually transitioning from one image to the next. This can create a smoother rotation effect in the animation. Here's an example of how to create a crossfade transition using the `ImageChops` module:

```python
from PIL import ImageChops

def crossfade(image1, image2, alpha):
    return ImageChops.blend(image1, image2, alpha)

crossfade_frames = []
for i in range(len(resized_earth_images) - 1):
    for alpha in (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9):
        frame = crossfade(resized_earth_images[i], resized_earth_images[i + 1], alp
        crossfade_frames.append(frame)
```

TRY IT

```
Command was successfully executed.
```

Click here to refresh your earth gif

In this code snippet, we create a `crossfade` function that blends two images using a specified `alpha` value. We then generate a list of crossfade frames by blending consecutive images in the `resized_earth_images` list.

2. Adding Visual Effects

You can also add visual effects to the animation, such as changing the brightness, contrast, or color of the images. In this example, we will adjust the brightness of the images using the ImageEnhance module:

```python
from PIL import ImageEnhance

def adjust_brightness(image, factor):
    enhancer = ImageEnhance.Brightness(image)
    return enhancer.enhance(factor)

brightness_factor = 1.5
brightened_frames = [adjust_brightness(frame, brightness_factor) for frame in cross
```

TRY IT

```
Command was successfully executed.
```

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In this code snippet, we create an `adjust_brightness` function that modifies the brightness of an image using a specified factor. We then create a list of brightened frames by applying the brightness adjustment to the `crossfade_frames` list.

3. **Creating the Enhanced Animation**
   Now that we have the crossfade transitions and visual effects in place, we can create an enhanced GIF animation using the modified frames:

```python
output_gif = "enhanced_rotating_earth.gif"
brightened_frames[0].save(
    output_gif,
    save_all=True,
    append_images=brightened_frames[1:],
    duration=duration,
    loop=loop_count,
    optimize=optimize
)
```

TRY IT

```
Command was successfully executed.
```

Click here to refresh your earth gif

With these transitions and visual effects, your animation should be more engaging and dynamic, showcasing the power and flexibility of the DALL-E 2 API.

GIF animation

Which of the following optimizations can be applied to a GIF animation to make it smoother and more visually appealing? (Select all that apply)

- [ ] Adjusting the frame duration
- [ ] Changing the loop count
- [ ] Adding crossfade transitions
- [ ] Applying visual effects like brightness and contrast adjustments

Adjusting the frame duration can make the animation appear faster or slower, allowing you to find the optimal speed for your animation.
Adding crossfade transitions between consecutive images can create a smoother rotation effect, enhancing the overall appearance of the animation.
Applying visual effects like brightness and contrast adjustments can make the animation more engaging and dynamic, showcasing the power and flexibility of the DALL-E 2 API.

Check It!